

NASDAQ stocks. Analysis of the skewness of absolute daily volumes finds NYSE = 4.33 and NASDAQ = 5.59, and large cap = 3.23 and small cap = 5.43. This shows that daily volumes are very positively skewed for all stocks (NYSE and NASDAQ, and large and small cap stocks). High kurtosis is also an indication of peaked means and fat tails, or in this case, an indication of outliers. Non-normal distribution patterns for asset returns and their repercussion on risk and return have been greatly researched and published. But the non-normal volume profiles and their repercussion on daily trading costs and ultimately portfolio returns have not been given the same level of attention as with asset prices. But it is just as important.

Another aspect of daily volume distribution and the use of “average” over “median” is that the positive skewness of the distribution causes the “average” volume metric to overestimate actual volumes the majority of times. Since volumes are highly skewed by positive outliers, it results in a value that could be much higher than the middle point of the distribution. In our data, the average is approximately in the 60–65% percentile, thus leading to an overestimation of daily volume 60–65% of the time. We have found (as we show in later chapters) that the median is a much more accurate predictor of daily volume than average. Why is this so? The reason is simple. When there are earnings announcements, company news, or macro events there will be more volume traded than on a normal day. Stock volume is highly skewed. As a fix, many practitioners compute the average by taking out the highest and lowest one or two data points, which results in a much more accurate measure. Interestingly, practitioners can compute a more accurate adjusted average by taking out the highest one or two data points since they are the most skewed outliers ([Table 2.9](#)).

Day of Week Effect

Historically there has always been a day of week pattern associated with market volumes. Stocks would trade the least amount on Monday, increasing on Tuesday and Wednesday and then decreasing on Thursday and Friday. Examination of large cap trading volume in 2012 found similar patterns. Mondays were the least active trading day with volume increasing on Tuesday–Thursday and falling slightly on Friday. We found similar patterns across NYSE and NASDAQ listed large cap stocks. Analysis of small cap stocks found a different pattern. Volumes were the lowest on Monday as expected, but highest on Tuesday and Friday. Similar patterns for small cap stocks were found for NYSE and NASDAQ listed stocks.

Next we examined the day of week effect for large and small cap stocks across time to see if the Friday effect for small cap stocks is relatively new